24

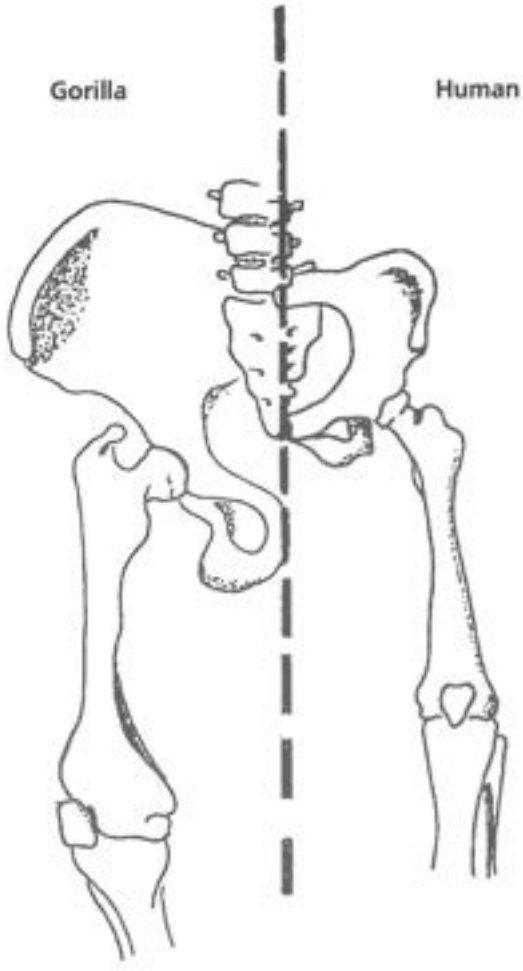
Homo sapiens

A species is made up of related organisms capable of interbreeding. They have a two-part name: the genus (first letter capitalised), then a Latin species name (no capitals). Scientists classify all humans currently living on Earth as members of a single species, *Homo sapiens*. The list below shows how we can classify *Homo sapiens* based on anatomical features:

- kingdom **Animalia** – an animal
- phylum **Chordata** – with a backbone
- subphylum **Vertebrata** – segmented spinal column
- class **Mammalia** – suckles its young
- subclass **Eutheria** – offspring develop with a placenta
- order **Primate** – hands/feet with five digits and two mammary glands on the chest
- suborder **Anthropoidea** – stereoscopic vision and a proportionately large brain
- family **Hominidae**

The primates that most closely resemble humans are chimpanzees, gorillas and orang-utans. Skeletons of *Homo sapiens* differ from these because of a very early adaptation to standing upright (erect posture) and walking on two feet (bipedalism). Humans have an S-shaped spinal column, putting the centre of gravity of the human body directly above the feet. This, and the modified pelvic region, makes walking upright easier.

Other anthropoids can walk upright to a limited extent, however, they all have spines which are straight or arched. They also have prehensile (grasping) feet and use their front limbs (hands) to lean on when walking. Because humans walk completely upright, hands have become much more developed, for example the thumb is fully opposable to the other fingers. Thus hands are able to make and manipulate tools. Tools were first made at least two and a half million years ago.



Modern humans have brains approximately twice the size of those of early human toolmakers. Babies have only a quarter of the brain capacity of adults and are dependent for a long time on parental care. During this time they learn and develop complex behaviour patterns. Full development of the human brain is thought to be complete at around 18 years of age.

The shape and size of human skulls have changed over time. Early human skulls had a sloping forehead and protruding jaws. Modern skulls have a high-rounded dome, straight-planed face, and smaller jaws. The ratio of brain size to body size is larger in humans than in primates. The cerebral cortex (covering of the brain) has become more developed, as are many other parts of the brain.

Estimates vary, but around 350 000 years ago, humans used fire to their benefit, wore clothes and organised hunting of large animals. The wearing of clothes enabled humans to move into cold areas, which would have otherwise been unhabitable for primates. It is likely that body hair was reduced in order to prevent overheating when hunting.

© Wendy St Germain, Peter Gribben and Learning Essentials 2003.

© Blake Education ISBN 978 1 92561 068 0 Assignments in Biology Book 3

The Stone Age began 30 000 to 40 000 years ago and Cro-magnon humans lived as hunters and gatherers. There is evidence of art, recording events and religion from this era. This implies that there must have been speech and conventions so that social groups required for such activities could exist. Some authorities argue that from about that time the genus *Homo* began to stabilise into the one species of *Homo sapiens*.

About 10 000 years ago the New Stone Age commenced. Pottery began to be made, crops were grown, village life and trading began. Cattle were domesticated and agriculture began to replace hunting in some areas. Many people think that a major factor in the success of humans was their domestication of plants and animals. Grasses were developed into wheat, barely and rice. Cattle and sheep were used to produce milk, clothing and meat.

1 If you have organisms which are very similar in appearance but cannot interbreed, are they of the same species? Explain your answer.

2 What is *bipedalism*?

3 What is the shape of the human spinal column?

4 What does *prehensile* mean?

5 Which three primates most closely resemble humans?

6 What development allowed humans to live in cold climates?

7 Describe the differences between modern and early human skulls?

8 What is an *anthropoid*?

9 What advantage is there for a baby to be cared for by parents for a long time?

10 Why do we think that *Cro-magnon* humans had language and a complex social system?

11 Why has it been an advantage for humans to be able to walk upright?

12 Discuss the following statement: *'Humans are unique in that they plan, transmit skills, have complex family groups, and modify the environment.'*

13 What is meant by *domestication*?

Complete the following passage using the following words:

five, live, mammals, milk, primates, teeth, temperature, thumbs

Humans are **14** _____ because they have a covering of fur or hair, have **15** _____ glands to suckle their young, have a constant body **16** _____(homeothermic) and give birth to **17** _____ young. Humans are **18** _____ because they have pentadactyl **19** (_____ – fingered) limbs, grasping hands, opposable **20** _____ well-developed binocular vision and unspecialised dentition **21** (_____).

© Wendy St Germain, Peter Gribben and Learning Essentials 2003.

© Blake Education ISBN 978 1 92561 068 0 Assignments in Biology Book 3